

# Seeing Green from Space: Machine Learning for Greenspace Detection

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# Workshop outline

## 1. Introduction to Satellite Data (15 mins)

applied in R

- ▶ Multispectral satellite instruments and images
- ▶ Key spectral indices, loading and visualising imagery

## 2. Machine Learning for Spatial Classification (15 mins)

theory

- ▶ Supervised vs. unsupervised ML for land use classification

## 3. Applied Land Use Classification (60 mins)

applied in R

- ▶ Classifying imagery into greenspace

# Imago: Make (satellite) imagery more useful, usable, and used

## Imagery Innovation



Ingesting pixels,  
producing  
metrics at scale

## Data for all



Meeting users  
where they are

## Capability & Community



Building a  
community  
around imagery

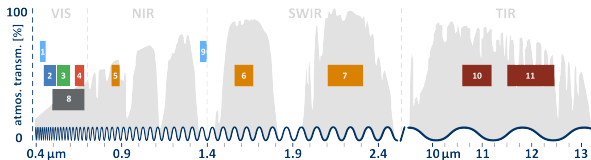


[imago.ac.uk](https://imago.ac.uk)

# Part 1: Introduction to Satellite Data

# What is a satellite image?

- ▶ Most products measure **reflectance**: the intensity of the sun's radiation reflected by the Earth's surface
- ▶ Reflectance is recorded across **different wavelengths** of the electromagnetic spectrum
- ▶ Because each place is measured at several wavelengths, one image holds multiple values per pixel, stored as separate raster layers
- ▶ These layers are called **bands**; the grid cells are **pixels**



Source: USGS; [blogs.fu-berlin.de/reseda/landsat-8](https://blogs.fu-berlin.de/reseda/landsat-8)

# Example: Landsat 8/9

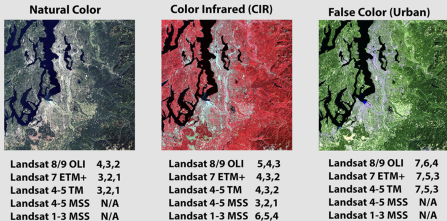
| Band | Name     | $\mu\text{m}$ | m   |
|------|----------|---------------|-----|
| 1    | aerosols | 0.44–0.45     | 30  |
| 2    | blue     | 0.45–0.51     | 30  |
| 3    | green    | 0.53–0.59     | 30  |
| 4    | red      | 0.64–0.67     | 30  |
| 5    | NIR      | 0.85–0.88     | 30  |
| 6    | SWIR-1   | 1.57–1.65     | 30  |
| 7    | SWIR-2   | 2.11–2.29     | 30  |
| 8    | pan      | 0.50–0.68     | 15  |
| 9    | cirrus   | 1.36–1.38     | 30  |
| 10   | TIR-1    | 10.6–11.2     | 100 |
| 11   | TIR-2    | 11.5–12.5     | 100 |

The same pixels combine into different **composites**:

- ▶ **Natural colour** (4,3,2): close to what the eye sees.
- ▶ **Colour infrared** (5,4,3): vegetation reflects NIR, so red marks healthy plants; water dark, urban white.
- ▶ **False colour, urban** (7,6,4): vegetation in greens (darker = denser); urban blue, soils brown.

## Common Landsat Band Combinations

Images: Landsat 8 Path 46 Row 27 acquired August 23, 2020. Band numbers displayed as R,G,B.



# Why satellite imagery, and why now?

## More satellites, more data

- ▶ Earth observation satellites grew  $\sim 700\%$  (2013–2023)
- ▶ Constellations now give near-daily revisits and sub-metre detail

## Better tools for big data

- ▶ Global archive  $\sim 800$  petabytes, growing  $\sim 100$  petabytes/year
- ▶ Cloud platforms, application programming interfaces, and cloud-native formats (Cloud-Optimised GeoTIFFs, SpatioTemporal Asset Catalog) [?]

# Different resolutions

Four kinds of **resolution** decide what we can observe:

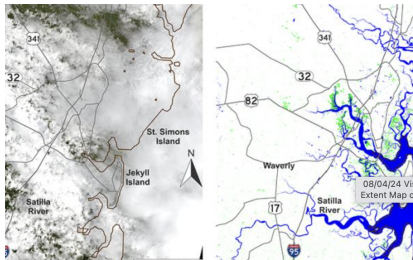
- ▶ **Spatial**: ground size of each pixel (level of detail)
- ▶ **Temporal**: how often a location is revisited
- ▶ **Spectral**: number and width of wavelength bands
- ▶ **Radiometric**: sensitivity to brightness differences

# Applications across social science

Satellite imagery now supports research and policy across many domains:

- ▶ **Land and environment:** land-use change, deforestation [?]
- ▶ **Urban:** settlement mapping, urban expansion, urban heat
- ▶ **Inequality:** greenspace access, environmental inequality
- ▶ **Disasters:** floods and wildfires (next slides)

# Application: near real-time flood mapping



Surface water (blue) and flood water (red).

Credit: NASA Worldview

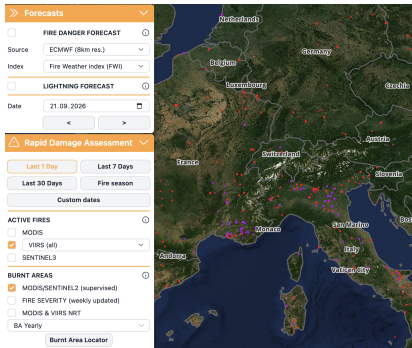
## NASA Global Flood Product

- ▶ Near real-time flood maps at ~250 m, within 3 hours of observation
- ▶ Built from MODIS and VIIRS sensors, delivered through NASA **LANCE**
- ▶ Used for disaster response and flood-risk assessment



[earthdata.nasa.gov/topics/human-dimensions/floods](https://earthdata.nasa.gov/topics/human-dimensions/floods)

# Application: wildfire monitoring



Copernicus EFFIS, Current Situation  
Viewer

## European Forest Fire Information System

- ▶ Active fires updated 6 times daily, within 2–3 hours of acquisition
- ▶ Burnt-area maps from Sentinel-2, MODIS, and VIIRS imagery
- ▶ Fire-danger forecasts up to 6 days ahead



[forest-fire.emergency.copernicus.eu](https://forest-fire.emergency.copernicus.eu)

# Multispectral imagery and spectral indices

Sensors measure many wavelengths beyond visible light, including infrared that the eye cannot see.

A **wavelength band** is a slice of the spectrum a sensor records (e.g. Sentinel-2's red band covers roughly 650–680 nm). Each band is one layer of information for the same place.

Combining bands separates surfaces that look alike to the eye: vegetation, soil, water, and built-up land (Sentinel-2 captures 13 bands).

A **spectral index** combines bands into a single measure. The most common is NDVI, which flags healthy vegetation:

$$\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}$$

**Hands-on:** load, visualise, and plot multispectral imagery.

## Part 2: Introduction to Machine Learning

# What is machine learning?

- ▶ Machine learning (ML) is a type of artificial intelligence (AI) that allows software applications to become more accurate at predicting outcomes without being explicitly programmed
- ▶ How does machine learning work? The original article is now behind a paywall. A YouTube video by the same author presents similar content. Relevant portion: 4:26 to 15:18 (minutes:seconds).

Useful general reference:

# How does the model “learn”?

## Unsupervised learning — no labels

- ▶ Finds patterns and structure within the data itself
- ▶ Tasks: clustering, dimensionality reduction, association rules

## Supervised learning — labelled data

- ▶ Input data paired with known output labels
- ▶ Learns an input  $\rightarrow$  output mapping to predict on unseen data
- ▶ Tasks: **classification** (categorical), **regression** (numerical)

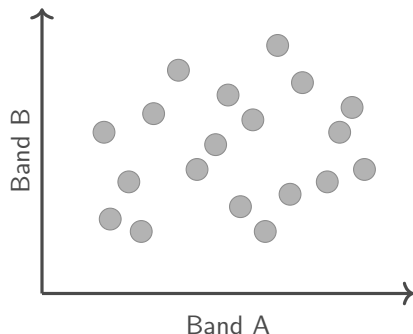
Either way, the model runs in two steps: **learning** (fit on training data) then **prediction** (apply to new data).

# Part 2A: Unsupervised Machine Learning

K-means clustering

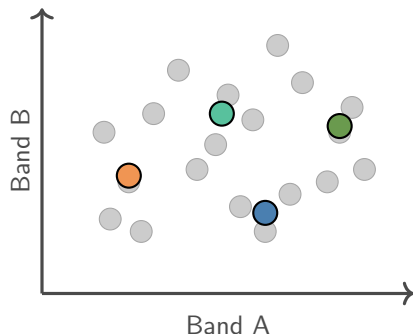
# K-means clustering (simplified)

- ▶ Each pixel is projected into **feature space**
- ▶ Every dimension (band) is a measurement (reflectance)
  - ▶ Usually many dimensions; 2 shown here



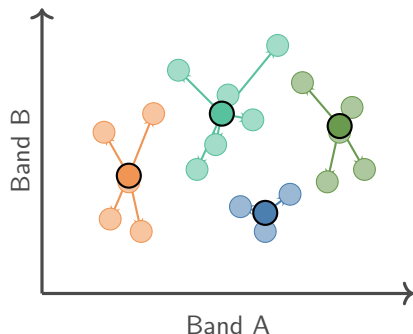
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- ▶ Pick **K random points** as class centres



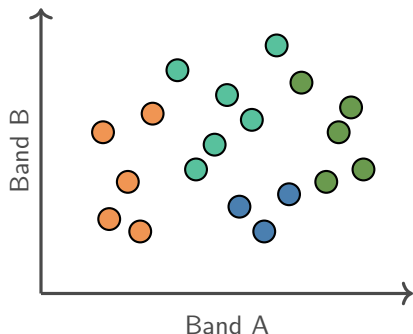
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- ▶ Group each point by distance to the closest centre



# K-means clustering (simplified)

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  - ▶ Usually many dimensions; 2 shown here
- ▶ Pick **K random points** as class centres
- ▶ Group each point by distance to the closest centre
- ▶ Assign a class to each point from its nearest centre



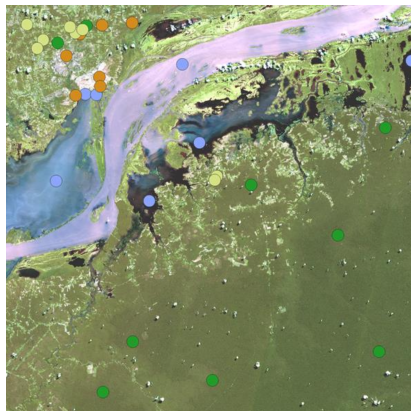
# Part 2B: Supervised Machine Learning

K-means clustering

# Classification: training data

## Training data can be:

- ▶ Points, lines, or polygons
- ▶ The pixel value(s) under each feature give a **spectral profile**
- ▶ This profile is how much light is reflected in the wavelengths of each band



NASA ARSET: Land Cover and Land Use Change

# Training data quality

## Good training data is:

- ▶ High-quality (accurate)
- ▶ Representative (varied)

## A model trained on good data:

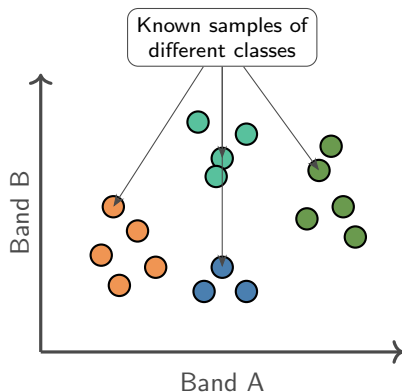
- ▶ Is robust to variation in the land cover itself (e.g. water turbidity)
- ▶ Is robust to errors between the land cover and the sensor (e.g. atmospheric effects)

`env_hazards_ineq/training_points`

NASA ARSET: Land Cover and Land Use  
Change

# Supervised classification: KNN overview

- ▶ **K-nearest neighbour** uses the labels of known samples (training data) to classify new ones
- ▶ K-means, by contrast, has no predefined classes; it only looks for similarity
- ▶ How it works:
  - ▶ For a new sample, check its K nearest known points
  - ▶ K is an integer  $> 1$  (odd avoids ties)
  - ▶ The majority vote wins

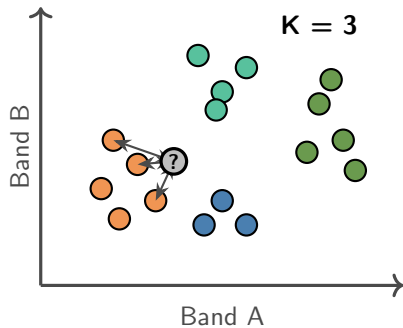


*"everything is related to everything else, but near things are more related than distant things" – Waldo Tobler*

# Supervised classification: KNN step-by-step

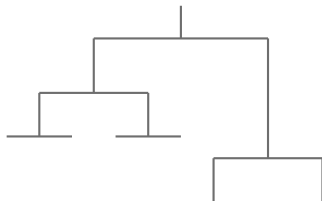
Watch how KNN classifies a new point:

- ▶ Set  $K = 3$
- ▶ Find the 3 closest known points
- ▶ Tally the votes
- ▶ Assign the new point to the winning class



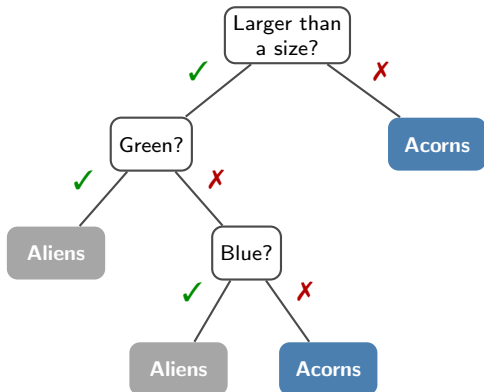
# Random forest: overview

- ▶ Random forest is a **tree-based** model
- ▶ It builds an array of decision trees
- ▶ Each tree offers a possible way to split the training data
- ▶ New data is classified by applying the tree(s)



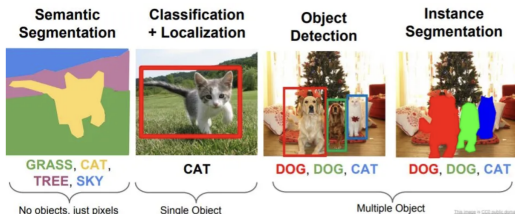
# Random forest: the decision tree

- ▶ Each split is a binary (true/false) question
- ▶ “Larger than some size?”
- ▶ “Is it green?”
- ▶ “Is it blue?”
- ▶ Aliens can be blue or green
- ▶ Acorns are smaller than aliens



# Supervised learning in images: what is the output?

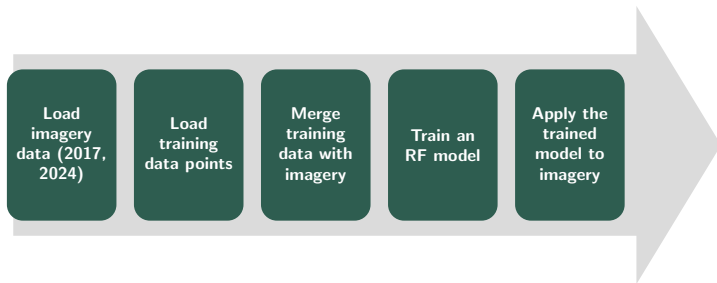
| Care about          | Want to know                | Method                |
|---------------------|-----------------------------|-----------------------|
| <b>Whole image</b>  | What kind of place is this? | Image classification  |
| <b>Object</b>       | Where is each thing?        | Object detection      |
| <b>Object shape</b> | What is its exact boundary? | Instance segmentation |
| <b>Pixel</b>        | What is at this location?   | Semantic segmentation |



Figures: [medium.com/inovako/object-detection-vs-image-segmentation-e5290e4690d](https://medium.com/inovako/object-detection-vs-image-segmentation-e5290e4690d)

# Building a random forest model in R

Roadmap for the following demo



# Get ready to code

1. Choose how you would like to follow the workshop:
  - ▶ **Run the exercises online via Binder**
  - ▶ Run the code locally in RStudio
  - ▶ Follow along without coding
2. Open the training website:  
`https://imago-sdruk.github.io/Imago\_training/`
3. Download the workshop data (website)
4. Code